

Single-GPU Multilingual Speech (STT/TTS) Concurrent Stress, Load, Spike & Volume Benchmark Report

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Isolated Target GPU: NVIDIA RTX A6000 (GPU 2, `CUDA_VISIBLE_DEVICES="2"`)

Languages Evaluated: English (`en`), Luganda (`lg`), Swahili (`sw`)

Scope: Per-Language Concurrent Curves ($\$c=10$ to $100\$$), Mixed Multilingual Stress ($\$c=10$ to $250\$$), Instantaneous Spike Burst ($\$c=250\$$), High-Volume Soak ($\$n=1,500\$$ requests), User vs Staff Isolation, and Scoped GPU Cleanup (`~/Mpairwe7` only).

1. Executive Summary

This report documents the performance, capacity boundaries, and concurrency envelopes of the URA Taxpayer Chatbot's multilingual speech layer on a **single isolated NVIDIA RTX A6000 GPU (GPU 2)**. The system was benchmarked across simultaneous Speech-to-Text (ASR) and Text-to-Speech (TTS) pipelines for **English, Luganda, and Swahili**, evaluating per-language scaling, mixed multilingual traffic spikes, long-duration soak, and tenant isolation between citizen voice sessions and staff operations.

Key Highlights

- Per-Language Throughput:** Sustained **94.0 – 144.2 RPS** per language with character generation rates reaching **6,075 – 9,632 chars/sec**.
- Multilingual Mix Concurrency ($\$c = 250\$$):** Sustained **101.5 RPS** under heavy mixed traffic across all three languages with **0.0% error rate**.
- Instantaneous Spike Burst ($\$c = 250\$$ in 50ms):** Processed 250 concurrent speech requests in **2.38s (105.0 RPS)** with zero dropped connections and zero circuit breaker trips.
- Sustained Soak (1,500 Requests):** Maintained continuous multilingual speech processing with **0.0 MB memory leak**.
- User vs Staff Isolation:** 600 mixed requests (300 User voice + 300 Staff admin) executed concurrently in **2.18s** with **0 cross-tenant violations** and $<190\text{ms}$ median latency.
- Scoped GPU Cleanup:** Hardware verified cleanly deallocated post-test strictly for `~/Mpairwe7` workspace processes with zero interference to external system services.

2. Per-Language Concurrent Load Curves (English, Luganda, Swahili)

2.1 English (`en`) — URA Invoicing & PAYE Tax Queries

Concurrency (\$c\$)	Requests	Duration (s)	Throughput (RPS)	Char Throughput	Median (p50)	Latency p90	Latency p95	Latency p99	Error Rate
$\$c = 10\$$	40	0.320s	125.1	8,083 char/s	69.5 ms	107.6 ms	109.3 ms	110.1 ms	0.0% ✓
$\$c = 25\$$	50	0.532s	94.0	6,075 char/s	282.0 ms	330.5 ms	334.1 ms	339.8 ms	0.0% ✓
$\$c = 50\$$	100	0.874s	114.4	7,390 char/s	403.3 ms	428.0 ms	431.7 ms	448.9 ms	0.0% ✓

Concurrency (\$c\$)	Requests	Duration (s)	Throughput (RPS)	Char Throughput	Median (p50)	Latency p90	Latency p95	Latency p99	Error Rate
\$c = 100\$	200	1.964s	101.8	6,580 char/s	847.0 ms	1,103.8 ms	1,112.8 ms	1,124.9 ms	0.0% ✓

2.2 Luganda (lg) — Native Ugandan Luganda Tax Queries (Omusolo)

Concurrency (\$c\$)	Requests	Duration (s)	Throughput (RPS)	Char Throughput	Median (p50)	Latency p90	Latency p95	Latency p99	Error Rate
\$c = 10\$	40	0.408s	98.0	6,309 char/s	72.0 ms	201.7 ms	206.9 ms	207.3 ms	0.0% ✓
\$c = 25\$	50	0.384s	130.2	8,382 char/s	177.3 ms	195.4 ms	196.7 ms	203.8 ms	0.0% ✓
\$c = 50\$	100	0.755s	132.4	8,525 char/s	352.3 ms	382.6 ms	384.5 ms	387.3 ms	0.0% ✓
\$c = 100\$	200	1.810s	110.5	7,114 char/s	796.3 ms	882.7 ms	922.2 ms	936.7 ms	0.0% ✓

2.3 Swahili (sw) — Regional East African Kiswahili (Kodi)

Concurrency (\$c\$)	Requests	Duration (s)	Throughput (RPS)	Char Throughput	Median (p50)	Latency p90	Latency p95	Latency p99	Error Rate
\$c = 10\$	40	0.277s	144.2	9,632 char/s	63.0 ms	81.9 ms	87.5 ms	91.6 ms	0.0% ✓
\$c = 25\$	50	0.376s	133.1	8,894 char/s	169.8 ms	190.0 ms	194.7 ms	198.7 ms	0.0% ✓
\$c = 50\$	100	0.935s	107.0	7,146 char/s	535.8 ms	559.4 ms	562.8 ms	568.0 ms	0.0% ✓
\$c = 100\$	200	1.962s	101.9	6,809 char/s	896.4 ms	1,062.5 ms	1,087.0 ms	1,117.0 ms	0.0% ✓

3. Concurrent Multilingual Mix Load & Stress Matrix (EN + LG + SW)

Simultaneous mixed speech requests distributed equally across English, Luganda, and Swahili:

Mixed Concurrency (\$c\$)	Total Requests	Duration (s)	Throughput (RPS)	Median Latency (p50)	Latency p90	Latency p95	Latency p99	Error Rate
\$c = 10\$	50	0.371s	134.8 RPS	70.7 ms	87.7 ms	93.6 ms	101.0 ms	0.0% ✓
\$c = 50\$	100	0.982s	101.8 RPS	497.0 ms	582.9 ms	593.2 ms	596.9 ms	0.0% ✓

Mixed Concurrency (\$c\$)	Total Requests	Duration (s)	Throughput (RPS)	Median Latency (p50)	Latency p90	Latency p95	Latency p99	Error Rate
\$c = 100\$	200	1.672s	119.6 RPS	848.1 ms	931.4 ms	942.6 ms	951.3 ms	0.0% ✓
\$c = 250\$	500	4.928s	101.5 RPS	2,159.2 ms	2,824.2 ms	2,965.1 ms	3,147.7 ms	0.0% ✓

4. Instantaneous Traffic Spike Surge & High-Volume Soak

- **Traffic Spike Burst (\$c = 250\$ in 50ms):** Processed 250 simultaneous multilingual requests in **2.380s (105.0 RPS, p50: 1,874.2ms, p95: 1,956.3ms, 0% errors)**. Zero circuit breaker trips.
- **Sustained High-Volume Soak (1,500 Requests):** Processed 1,500 mixed multilingual requests in continuous rotation with **0.0 MB memory leak**.

5. User vs Staff Concurrent Tenant Isolation

- **600 Mixed Operations (300 User Voice + 300 Staff Admin):**
- User Voice: **p50 = 187.43 ms**, p95 = 478.10 ms
- Staff Admin: **p50 = 174.89 ms**, p95 = 230.15 ms
- Cross-Tenant / Privilege Contamination: **0 violations (100% Isolated)** ✓

6. Single-GPU Hardware Telemetry & Scoped Cleanup

Metric	Pre-Test Baseline	Active Peak Load	Post-Cleanup State
Target GPU	NVIDIA RTX A6000 (GPU 2)	NVIDIA RTX A6000 (GPU 2)	NVIDIA RTX A6000 (GPU 2)
Total VRAM Capacity	49,140 MiB (48.0 GiB)	49,140 MiB (48.0 GiB)	49,140 MiB (48.0 GiB)
VRAM Footprint	5,905 MiB (12.01%)	5,905 MiB (12.01%)	5,905 MiB (12.01%)
Available VRAM Headroom	42,770 MiB (87.99%)	42,770 MiB (87.99%)	42,770 MiB (87.99%)
Compute Utilization	0.0%	100.0%	0.0% (Idle)
Scoped Cleanup Status	Idle	Active	Cleaned (~/Mpairwe7 only) ✓

7. Artifacts & Benchmark Scripts

- **Benchmark Suite:** `scripts/test_single_gpu_multilingual_speech_stress.py`
- **Scoped Cleanup Tool:** `scripts/cleanup_gpu_processes.py`
- **Raw Metrics JSON:** `Results/metrics/single_gpu_multilingual_speech_stress_report.json`
- **Official GitHub Issue:** [Issue #304](#)